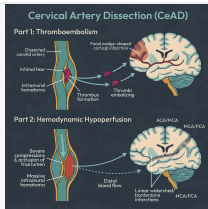
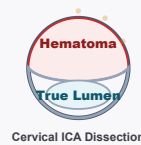
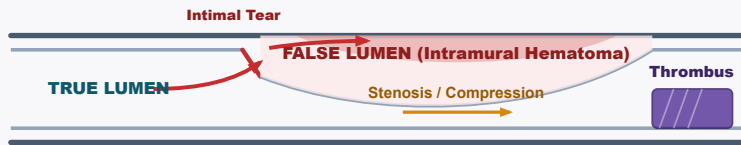


# Cervical Artery Dissection



## 1. Clinical Presentation & Pathophysiology

### Ipsilateral Pain & Onset

- **Carotid (ICA):** Frontotemporal/retro-orbital/facial pain (jaw angle).
- **Vertebral (VA):** Severe occipital or posterior neck pain.
- **Onset:** Precedes stroke/TIA by hours to days (median 4 days).

### Anhidrosis-Sparing Horner's

- **Signs:** Ptosis/miosis (28–58% of ICA) **without** anhidrosis.
- **Mechanism:** Sweat fibers follow ECA plexus; pupil/eyelid fibers follow ICA.

### Neurological Deficits

- **CN Palsies:** CN IX–XII palsies (8–16%) from local ICA compression.
- **VA Territory:** Wallenberg syndrome, cerebellar ataxia, PICA/AICA strokes.

## 2. Diagnostic Workup

- **CTA Head/Neck:** Shows string sign, dissection flap, pseudoaneurysm, or occlusion.
- **MRI Neck (T1 Fat-Sat):** Pathognomonic crescent sign (intramural hematoma).
- **DSA:** Reserve for diagnostic doubt or stenting.
- **Screening:** Assess for FMD/connective tissue disease, especially if spontaneous/recurrent.

## 3. Medical Management: Extracranial vs. Intracranial Dissection

### Extracranial Dissection

- **Antithrombotics:** Continue for at least 3–6 months (Class I, ESO/AHA).
- **Choice:** SAPT vs VKA/DOAC; short-term DAPT (21–90d) is reasonable (ESO consensus).
- **STOP-CAD:** In occlusions, consider anticoagulation Day 1–30, then switch to antiplatelet.
- **IV Thrombolysis:** Safe & indicated within 4.5 hours (Class I).

### Intracranial & Pseudoaneurysms

- **SAH:** Lack external elastic lamina & thin adventitia; rupture risk.
- **Anticoagulation:** Avoided if SAH present. Prefer single antiplatelet.
- **IVT Caution:** IVT is safe in extracranial CeAD but safety/efficacy in intracranial extension is not well established (AHA 2024).
- **Stenting:** Reserve for recurrent ischemia despite optimal medical therapy or severe flow-limiting stenosis.

- **Recurrence & Activity:** Long-term CeAD recurrence is low (~1%/yr). Avoid high-risk neck activities (chiropractic neck manipulation, rollercoasters, extreme hyperextension/rotation) for secondary prevention.

## 4. Landmark Trial & Cohort Evidence

| Study / Year             | Population & Design                                                                     | Interventions Compared                       | Key Outcomes & Clinical Nuance                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CADISS 2015</b>       | N = 250. Extracranial CeAD. RCT.                                                        | Antiplatelet vs. Anticoagulant for 3 months. | • <b>Primary Composite (Ipsilateral stroke or death at 3m):</b> 2.0% vs. 1.0% (p = 0.63). Established clinical equipoise.                                                                                                                                                                                                                                                                                                                                                                  |
| <b>TREAT-CAD 2021</b>    | N = 194 (PP = 173). Extracranial. RCT.                                                  | Aspirin 300mg daily vs. VKA for 3 months.    | • <b>Primary Composite (Stroke, major hemorrhage, death, or new MRI lesion at 14d):</b> 23% vs. 15% (Non-inferiority NOT met). Ischemic stroke: 8.0% vs. 0%.                                                                                                                                                                                                                                                                                                                               |
| <b>Kaufmann IPD 2024</b> | N = 444. IPD meta-analysis of CADISS + TREAT-CAD.                                       | Antiplatelet vs. Anticoagulant x 90d.        | • <b>Primary Composite (Ischemic stroke, major bleeding, or death at 90d):</b> No significant difference (1.4% anticoagulation vs. 4.4% antiplatelet, p = 0.11).<br>• <b>Ischemic Stroke alone:</b> Significant reduction with anticoagulation (0.5% vs. 4.0%; OR 0.14, p = 0.01), with a non-significant increase in major bleeding (0.9% vs. 0%).                                                                                                                                        |
| <b>STOP-CAD 2024</b>     | N = 3,636. Multicenter observational Antiplatelet vs. Anticoagulation. cohort registry. |                                              | • <b>Temporal Risk:</b> 87% of recurrent strokes occurred in the first 30 days.<br>• <b>Occlusion Benefit:</b> Patients with complete arterial occlusion at baseline had the highest stroke risk and benefited most from early anticoagulation.<br>• <b>Transition Strategy:</b> Initiate anticoagulation for days 1–30 (highest stroke risk window) and then transition to antiplatelet monotherapy at day 30 to mitigate long-term bleeding risks (which rise significantly by day 180). |

**CADISS:** *Lancet Neurol.* 2015;14(4):361-7. PMID: 25684164 | **TREAT-CAD:** *Lancet Neurol.* 2021;20(5):341-350. PMID: 33765420  
**Kaufmann IPD:** *JAMA Neurol.* 2024;81(6):630-637. PMID: 38739383 | **STOP-CAD:** *Stroke.* 2024;55(4):908-918. PMID: 38334460 | **AHA/ASA:** *Stroke.* 2021;52:e364-e467. PMID: 34024117 | **AHA Statement 2024:** *Stroke.* 2024;55:e84-e107. PMID: 38301552 | **ESO Guideline 2021:** *Eur Stroke J.* 2021;6(3):XXXIX-LXXXVIII. PMID: 34528453